

Angki Myardi
1102220019

No.

Date

Tugas Minggu 2

1. Ukuran File

dimension = 300×400

A. Grayscale

ukuran per pixel = 1 byte

ukuran file = $300 \times 400 \times 1 = 120000 \text{ byte} = \frac{120.000}{1024} \text{ KB}$

= 117,1875 KB

B. Color 16-bit

ukuran per pixel = 16-bit = 2 byte

ukuran file = $300 \times 400 \times 2 = 240000 \text{ byte} = \frac{240.000}{1024} \text{ KB}$

= 234,375 KB

C. color 24-bit

ukuran per pixel = 24-bit = 3 byte

ukuran file = $300 \times 400 \times 3 = 360000 \text{ byte} = \frac{360.000}{1024} \text{ KB}$

= 351,5625 KB

2. Ukuran File

4x6 inch dengan dpi = 600

dimensi = $4 \text{ dpi} \times 6 \text{ dpi} = 24 \text{ dpi}^2$

= $24 \times 600^2 = 8.640.000 \text{ pixel}$

ukuran per pixel = 3 byte

ukuran file = $8 \times 640.000 \times 3 = 25.920.000 \text{ byte}$

kompresi 5:3 =

$25.920.000 \times \frac{3}{5} = 15.552.000 \text{ byte}$

$\frac{15.552.000}{1024} \text{ KB} = 15107,5 \text{ KB}$

$\frac{15107,5}{1024} \text{ MB} = 14,83 \text{ MB}$

3. a. Kebutuhan bandwidth minimum :

- video

$$\begin{aligned}\text{Resolusi per frame} &= 352 \times 288 \\ &= 101.376 \text{ px}\end{aligned}$$

$$\text{True color} = 101.376 \times 24 \text{ bits}$$

$$= 2.433.024 \text{ bits}$$

$$\text{ukuran} = 2.433.024 \times 25 \times 1$$

$$= 60.825.600 \text{ bit/sec}$$

$$= \frac{60.825.600}{1000.000}$$

$$= 60,8256 \text{ mbps}$$

- Audio

$$\text{Sampling rate} = 44 \text{ kHz}$$

$$= 44.100 \text{ Hz}$$

$$\text{Penyimpanan} = 16 \text{ bit/sampel}$$

$$\text{ukuran} = 44.100 \times 16 \times 2$$

$$= 1.411.200 \text{ bit/sec}$$

$$= \frac{1.411.200}{1000.000}$$

$$= 1,4112 \text{ mbps}$$

$$\text{Bandwidth minimum} = 60,841,41$$

$$= 62,21 \text{ mbps}$$

b. waktu yang dibutuhkan untuk 20 frame dengan 10Mbps ethernet:

$$\begin{aligned}\text{ukuran: } & 2,433,024 \times 20 \times 1 \\ & = 48.660.480 \text{ bit/sec} \\ & = 48.660.480 / 1000.000 \\ & = 48.66 \text{ mbps}\end{aligned}$$

waktu yang dibutuhkan ~~untuk~~ jika tanpa audio

$$\begin{aligned}& = 48.66 \text{ mbps} \\ & \quad \underline{10 \text{ mbps}} \\ & = 4.866 \text{ sec}\end{aligned}$$

waktu yang dibutuhkan jika menggunakan audio

$$\begin{aligned}& = (48.66 + 1.412) \text{ mbps} / 10 \text{ mbps} \\ & = 50.0712 \\ & \quad \underline{10 \text{ mbps}} \\ & = 5.00712 \text{ seconds}\end{aligned}$$

c. minimum rasio kompleks

rasio = bandwidth min / kapasitas Ethernet

$$\begin{aligned}& = \frac{62.21}{10 \text{ mbps}} \\ & = 6.221\end{aligned}$$

$$\begin{aligned}\text{Rasio presentase} &= 1 - (\text{Kapasitas Ethernet} / \text{Bandwidth min}) \times 100\% \\ &= (1 - \frac{10}{62.21}) \times 100\% \\ &= 83.91\%\end{aligned}$$